

Kiến trúc và thiết kế phần mềm

Lai Thanh Sĩ - 22639301

Tuần 01

1. Chạy docker-compose: `docker-compose -f docker-compose.yml up`

```
○ → rabbitmq git:(main) × docker-compose -f docker-compose.yml up
[+] Running 4/4
  ✓ Network rabbitmq_default      Created           0.0s
  ✓ Container rabbitmq-python     Created           0.0s
  ✓ Container consumer-python     Created           0.1s
  ✓ Container producer-python     Created           0.1s
Attaching to consumer-python, producer-python, rabbitmq-python
consumer-python | Consumer connecting...
consumer-python | Consumer failed, retry in 3s...
producer-python | Waiting for RabbitMQ...
rabbitmq-python | 2026-01-24 13:00:00.157573+00:00 [notice] <0.45.0> Application syslog exited with reason: stopped
rabbitmq-python | 2026-01-24 13:00:00.161109+00:00 [notice] <0.209.0> Logging: switching to configured handler(s); following
messages may not be visible in this log output
rabbitmq-python | 2026-01-24 13:00:00.161404+00:00 [notice] <0.209.0> Logging: configured log handlers are now ACTIVE
rabbitmq-python | 2026-01-24 13:00:00.166643+00:00 [info] <0.209.0> ra: starting system coordination
rabbitmq-python | 2026-01-24 13:00:00.166673+00:00 [info] <0.209.0> starting Ra system: coordination in directory: /var/lib/r
abbitmq/mnesia/rabbit@ba91172d6c7b/coordination/rabbit@ba91172d6c7b
rabbitmq-python | 2026-01-24 13:00:00.172188+00:00 [info] <0.216.0> ra_coordination_log_ets: in system coordination initialis
ing. Mem table opts: [set,{write_concurrency,auto},public,{compressed,false}]
rabbitmq-python | 2026-01-24 13:00:00.202508+00:00 [info] <0.222.0> ra system 'coordination' running pre init for 0 registere
d servers
rabbitmq-python | 2026-01-24 13:00:00.208661+00:00 [info] <0.223.0> ra: meta data store initialised for system coordination.
0 record(s) recovered
rabbitmq-python | 2026-01-24 13:00:00.211488+00:00 [info] <0.226.0> segment_writer: upgrading segment file names to new forma
t in directory /var/lib/rabbitmq/mnesia/rabbit@ba91172d6c7b/coordination/rabbit@ba91172d6c7b
rabbitmq-python | 2026-01-24 13:00:00.220723+00:00 [notice] <0.228.0> WAL: ra_coordination_log_wal init, mem-tables table nam
e: ra_coordination_log_open_mem_tables
rabbitmq-python | 2026-01-24 13:00:00.225774+00:00 [info] <0.209.0> ra: starting system quorum_queues
rabbitmq-python | 2026-01-24 13:00:00.225804+00:00 [info] <0.209.0> starting Ra system: quorum_queues in directory: /var/lib/r
abbitmq/mnesia/rabbit@ba91172d6c7b/quorum/rabbit@ba91172d6c7b
rabbitmq-python | 2026-01-24 13:00:00.226177+00:00 [info] <0.232.0> ra_log_ets: in system quorum_queues initialising. Mem tab
le opts: [set,{write_concurrency,auto},public,{compressed,true}]
rabbitmq-python | 2026-01-24 13:00:00.226797+00:00 [info] <0.236.0> ra system 'quorum_queues' running pre init for 0 register
ed servers
rabbitmq-python | 2026-01-24 13:00:00.227150+00:00 [info] <0.237.0> ra: meta data store initialised for system quorum_queues.
0 record(s) recovered
rabbitmq-python | 2026-01-24 13:00:00.227230+00:00 [info] <0.240.0> segment_writer: upgrading segment file names to new forma
t in directory /var/lib/rabbitmq/mnesia/rabbit@ba91172d6c7b/quorum/rabbit@ba91172d6c7b
rabbitmq-python | 2026-01-24 13:00:00.227971+00:00 [notice] <0.242.0> WAL: ra_log_wal init, mem-tables table name: ra_log_ope
n_mem_tables
rabbitmq-python | 2026-01-24 13:00:00.228838+00:00 [info] <0.244.0> ra_system_recover: ra system 'quorum_queues' server recov
ery strategy rabbit_quorum_queue:system_recover
rabbitmq-python | 2026-01-24 13:00:00.228879+00:00 [info] <0.244.0> [rabbit_quorum_queue:system_recover/1] rabbit not booted,
skipping queue recovery
rabbitmq-python | 2026-01-24 13:00:00.228937+00:00 [info] <0.209.0> ra: starting system coordination
rabbitmq-python | 2026-01-24 13:00:00.228947+00:00 [info] <0.209.0> starting Ra system: coordination in directory: /var/lib/r
abbitmq/mnesia/rabbit@ba91172d6c7b/coordination/rabbit@ba91172d6c7b
rabbitmq-python | 2026-01-24 13:00:00.285797+00:00 [notice] <0.246.0> RabbitMQ metadata store: candidate -> leader in term: 1
machine version: 2, last applied 0
```

```

Feature `management_metrics_collection` is deprecated.
rabbitmq-python | 2026-01-24 13:00:01.371906+00:00 [warning] <0.475.0> By default, this feature can still be used for now.
rabbitmq-python | 2026-01-24 13:00:01.371906+00:00 [warning] <0.475.0> Its use will not be permitted by default in a future m
minor RabbitMQ version and the feature will be removed from a future major RabbitMQ version; actual versions to be determined.
rabbitmq-python | 2026-01-24 13:00:01.371906+00:00 [warning] <0.475.0> To continue using this feature when it is not permitte
d by default, set the following parameter in your configuration:
rabbitmq-python | 2026-01-24 13:00:01.371906+00:00 [warning] <0.475.0> "deprecated_features.permit.management_metrics_col
lection = true"
rabbitmq-python | 2026-01-24 13:00:01.371906+00:00 [warning] <0.475.0> To test RabbitMQ as if the feature was removed, set th
is in your configuration:
rabbitmq-python | 2026-01-24 13:00:01.371906+00:00 [warning] <0.475.0> "deprecated_features.permit.management_metrics_col
lection = false"
rabbitmq-python | 2026-01-24 13:00:01.390249+00:00 [info] <0.511.0> Management plugin: HTTP (non-TLS) listener started on por
t 15672
rabbitmq-python | 2026-01-24 13:00:01.390360+00:00 [info] <0.539.0> Statistics database started.
rabbitmq-python | 2026-01-24 13:00:01.390407+00:00 [info] <0.538.0> Starting worker pool 'management_worker_pool' with 3 proc
esses in it
rabbitmq-python | 2026-01-24 13:00:01.396242+00:00 [info] <0.550.0> Prometheus metrics: HTTP (non-TLS) listener started on po
rt 15692
rabbitmq-python | 2026-01-24 13:00:01.396287+00:00 [info] <0.453.0> Ready to start client connection listeners
rabbitmq-python | 2026-01-24 13:00:01.396933+00:00 [info] <0.594.0> started TCP listener on [::]:5672
rabbitmq-python | 2026-01-24 13:00:01.455404+00:00 [info] <0.453.0> Server startup complete; 4 plugins started.
rabbitmq-python | 2026-01-24 13:00:01.455404+00:00 [info] <0.453.0> * rabbitmq_prometheus
rabbitmq-python | 2026-01-24 13:00:01.455404+00:00 [info] <0.453.0> * rabbitmq_management
rabbitmq-python | 2026-01-24 13:00:01.455404+00:00 [info] <0.453.0> * rabbitmq_management_agent
rabbitmq-python | 2026-01-24 13:00:01.455404+00:00 [info] <0.453.0> * rabbitmq_web_dispatch
rabbitmq-python | completed with 4 plugins.
rabbitmq-python | 2026-01-24 13:00:01.498386+00:00 [info] <0.10.0> Time to start RabbitMQ: 2726 ms
consumer-python | Consumer connecting...
rabbitmq-python | 2026-01-24 13:00:01.760798+00:00 [info] <0.599.0> accepting AMQP connection 172.19.0.3:42606 -> 172.19.0.2:
5672
rabbitmq-python | 2026-01-24 13:00:01.772571+00:00 [info] <0.599.0> connection 172.19.0.3:42606 -> 172.19.0.2:5672: user 'use
r' authenticated and granted access to vhost '/'
consumer-python | Waiting for messages...
rabbitmq-python | 2026-01-24 13:00:01.804756+00:00 [info] <0.620.0> accepting AMQP connection 172.19.0.4:34638 -> 172.19.0.2:
5672
rabbitmq-python | 2026-01-24 13:00:01.806742+00:00 [info] <0.620.0> connection 172.19.0.4:34638 -> 172.19.0.2:5672: user 'use
r' authenticated and granted access to vhost '/'
producer-python | Producer connected to RabbitMQ
producer-python | * Serving Flask app 'producer_d1q'
producer-python | * Debug mode: off
producer-python | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI serv
er instead.
producer-python | * Running on all addresses (0.0.0.0)
producer-python | * Running on http://127.0.0.1:3000
producer-python | * Running on http://172.19.0.4:3000
producer-python | Press CTRL+C to quit

```

[View in Docker Desktop](#)
[View Config](#)
[Enable Watch](#)

2. Gửi request để kiểm tra

```

● → rabbitmq git:(main) x curl -X POST http://localhost:3000/send \
-H "Content-Type: application/json" \
-d '{"message": "Test message", "orderId": "Order #1000"}'
{"dataSent": {"message": "Test message", "orderId": "Order #1000", "timestamp": "2
026-01-24T13:01:33.354834"}, "status": "sent"}
○ → rabbitmq git:(main) x █

```

```

producer-python | Sent: {'message': 'Test message', 'orderId': 'Order #1000
', 'timestamp': '2026-01-24T13:01:33.354834'}
producer-python | 192.168.65.1 -- [24/Jan/2026 13:01:33] "POST /send HTTP/
1.1" 200 -
consumer-python | Processing: {"message": "Test message", "orderId": "Order
#1000", "timestamp": "2026-01-24T13:01:33.354834"}
consumer-python | Process success
█

```

